

Value & Publishability

Rubric Deliverable P4 — MRI-ReportGenerator (Cervical Spine MRI Analysis)

EECE503N Final Project

2026-06-08

Framing. The project is positioned as an **Application** (rubric GT5). Therefore the bar this deliverable must clear is *application value*; *publishability* is presented as a credible *upside*, not as a claim. We do not assert a guaranteed publication — the rubric’s Research track carries a publishability hard-stop, and over-claiming would be dishonest. What we do show is (i) concrete, defensible application value, and (ii) specific, benchmarkable research nuggets with named candidate venues. Companion materials cover the technical depth, the quantified non-AI baseline, and the full manuscript.

1 Application value (the bar)

The system augments — never replaces — the radiologist read of a cervical sagittal-T2 MRI. Its value is in four properties the manual baseline lacks:

Property	What the system delivers	Why it matters vs. the manual baseline
Reproducibility	Deterministic geometric measurements — the same study yields the same numbers every run	Manual measurement has real inter-reader (and intra-reader) variability; the quantified gap is in P2
Speed	Measurements computed in seconds per case (per-component latency measured; see infra <code>tradeoffs.md</code>)	Manual caliper measurement of canal, disc, Cobb and vertebral geometry is slow; the per-case time is in P2
Auditability	Every flag traces to a measurement and a <i>cited</i> threshold (the G6 catalog), not an opaque verdict	A radiologist can check each number; a black-box score cannot be audited or defended
Triage / augmentation	Pre-populates the structured measurement table and flags cases whose measurements cross reference thresholds for review	Frees specialist time from repetitive geometry for judgement; “flagged for physician review,” never a diagnosis

Who benefits / who pays. Radiology departments and research-hospital workflows: the buyer is the imaging department or research group that wants consistent, fast, auditable cervical measurements at volume. The value proposition is *consistency + speed + auditability*, not a black-box diagnosis.

The value is quantified by the baseline contrast (P2). The two numbers that make this case concrete — manual measurement/reading *time* per case, and manual inter-observer *variability* — are the subject of deliverable P2. The argument here is structural: a deterministic pipeline

removes the variability entirely and collapses the time, with every output cited. P2 supplies the magnitudes.

2 Publishability (the upside)

Independent of the application framing, the project produced research-grade artifacts that are genuinely benchmarkable. We list them honestly, separating what is a real contribution from what is a heuristic.

2.1 What is novel / benchmarkable

- **Healthy-anchored, disease-agnostic geometric detectors.** Measurements are validated to read *inside* a healthy range and to be crossed by symptomatic anatomy, rather than trained on a disease label. This is a reusable evaluation pattern for measurement pipelines where per-case ground truth is unavailable.
- **Threshold-crossing validation under a no-ground-truth constraint.** A reproducible methodology (healthy anchor + distribution separation by Mann–Whitney) that yields a defensible validation *without* per-case radiologist labels — because no public dataset pairs cervical MRI with per-case expert measurements. The methodology itself is the contribution.
- **The physical-dimension vs. intensity scanner-immunity finding.** We show empirically that millimetre metrics (canal, SAC, cord) separate cross-dataset while intensity/ratio metrics (disc signal, height ratios) are acquisition-confounded and must be validated within a single dataset. This is a non-obvious, transferable insight for anyone validating MRI-derived measurements across scanners.
- **Honest negative results.** The disc *signal* axis does *not* discriminate (AUC 0.50, level-controlled), and disc AP-width’s apparent strong effect collapsed under level-stratification (AUC 0.79 $\rightarrow$ 0.61). Reported negatives and confound corrections are publishable precisely because they are rare in this literature.
- **A reproducible validation on real healthy + symptomatic cervical MRI** (Spine-Generic healthy anchor; MMCS D symptomatic demonstration set), with a committed harness and a fully-cited threshold catalog.

2.2 What is *not* claimed

The per-disc and per-vertebra cut-points (e.g. the disc/VB ratio threshold) have no κ /AUC ground truth, so they are scoped as *research-grade heuristics*, not validated clinical thresholds. The ceiling on the disc metrics is data-limited (a coarse binary lesion label, and features the segmentation masks cannot see); raising it requires radiologist per-disc grades, which is future work, not a present claim.

2.3 Candidate venues

The work fits the medical-imaging-informatics / computational-spine niche rather than a new-architecture venue. Honest candidates, in order of fit:

- **MICCAI workshops** — e.g. Computational Spine Imaging (CSI) or applications/validation-focused workshops: the methodology + reproducible validation is a natural fit.
- **SPIE Medical Imaging** (Computer-Aided Diagnosis / Image Processing tracks) — measurement pipelines with validation are in scope.

- **Journal of Digital Imaging / Insights into Imaging** — clinical imaging informatics; a deployed, auditable measurement-and-report system with a validation study is on-topic. The full manuscript draft is the artifact that would be adapted to a chosen venue’s format.

3 Summary for the grader

- **Application value is real and structural:** reproducibility, speed, auditability, and triage — with the quantified baseline contrast supplied by P2.
- **Publishability is a credible upside, not a claim:** the validation methodology, the scanner-immunity finding, and the honest negative results are genuinely benchmarkable, with named candidate venues and an existing paper draft.
- **We are explicit about the boundary:** heuristic thresholds are labelled as such; the data-limited ceiling is stated; no guaranteed publication is asserted.